

DETERMINATION AND COMPARISON OF VITAMIN C CONTENT FROM MORINGA OLEIFERA BY DIFFERENT METHODS

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ABSTRACT

Moringa oleifera is a perfect example of multipurpose tree. Moringa is an important food commodity which has had attention as natural nutrition of the tropics. Moringa contains vitamin A, B, good source of vitamin C (ascorbic acid), palmitic acid, steric acid, calcium and potassium. Its seeds contain high quality edible oil (up to 40% by weight) and water soluble proteins that act as effective coagulants for water and wastewater treatment. Stem, fresh leaves and dried leaves samples of Moringa have been analyzed for ascorbic acid content (Vitamin C) using titration, enzymatic and HPLC methods. Fresh leaves of Moringa were found to be rich source of Vitamin C when compared with lemon and mosambi. Fresh leaves of Moringa was found to contain ~0.82mg/g Vitamin C which is roughly 2-4 times higher than that of from known sources of Vitamin C viz. lemon and mosambi.

KEYWORDS: Moringa Oleifera, Vitamin C, Quantification, Titration, Spectrophotometer, HPLC

Abbreviations: mg- Milligram, US- United States, HPLC - High Performance Liquid Chromatography, g- Grams, Vit C- Vitamin C

INTRODUCTION

Moringa oleifera L. is a tropical plant belonging to one of the 14 species of family moringaceae (Iqbal and Bhangar, 2006). The tree has its origin from Agra and Oudh in northern eastern region of India, in south of Himalayas and is now cultivated across whole of tropical belt (Mughal et al., 1999). It is referred by many names such as drumstick tree, horseradish tree, never die tree and miracle tree (Ramachandran et al., 1980). Moringa has been reported to provide human, livestock and many crop nutritional benefits (Fuglie, 2001).

Different parts of Moringa plant have a number of medicinal properties which have been recognized by both Ayurvedic and Unani systems of medicines (Mughal et al., 1999). The leaves of M. oleifera have been reported to be a valuable source of both macro and micronutrients, rich source of β -carotene, protein, calcium, potassium and act as a good source of natural antioxidants such as ascorbic acid, flavonoides, phenolics and carotenoides (Fuglie 1999; Dillard and German 2000; Siddhuraju and Becker, 2003). The important components of Moringa like calcium and vitamin C found to be suitable for exploitation by many developing regions of world where malnutrition is a major concern, specially for the children (Fahey, 2005).

Diet containing vitamins help to maintain health of humans. Vitamin C is essential micro-nutrient required for normal metabolic functions of human body (Jaffe 1984), supplied by fruits and vegetables. According to US standards

minimum of 60 mg of ascorbic acid is a daily requirement by a human body (Ting, 1980). Moringa leaves contain more vitamin A than carrots, more vitamin C than oranges, more calcium than milk, more potassium than bananas and more iron than spinach (Fahey, 2005). Ligaments, tendons and collagen rely on vitamin C to stay strong and healthy. It has been shown by different research that a passable intake of vitamin C is effective in lowering the risk of developing cancers of mouth, lung, breast, stomach, cervix, colon, rectum and prostate (Levine et al., 1996; Block 1992; Frei 1994; Jacobs 1993).

As a potent antioxidant, vitamin C has capacity to eliminate several different reactive oxygen species, keeps membrane bound anti oxidant α -tocopherol in reduced state, appears to be substrate for oxalate and tartrate biosynthesis and has a role in stress resistance (Arrigoni and De Tullio 2002; Davey et al., 2000; Klein and Kurilich, 2000). It is used in cure of certain diseases such as scurvy (Levine, 1986), common cold (Hemila, 1992), anemia and even infertility. As a result of mutation in pseudogene coding for enzyme (L- gulonolactone oxidase) required for bio-synthesis of vitamin C via glucuronic acid pathway, humans and other primates have lost their ability to synthesize vitamin C (Woodall and Arnes, 1997). Thus, vitamin C should be obtained through diet. Since Moringa is good source of vitamin C yet its use as a good source of ascorbic acid is not explored much.

Keeping in view the importance of Moringa plant, vitamin C was quantified in leaves (dry and fresh) and stem using three independent methods viz. titration, spectrophotometric and HPLC. Moringa leaves were collected from Solan district of Himachal Pradesh. Three methods were used as per the previous literature available for determination of Vitamin C are titration (Hassan and Hassan, 2008), spectrophotometer (Raghu et al., 2007) and High performance liquid chromatography (HPLC) (Raghu et al., 2007; Burini, 2007). A part of the leaves were dried and both fresh and dried leaves and stem as samples were used for determination of vitamin C. For the preparation of standard curve, Ascorbic acid of analytical grade (SIGMA) was used.

Lemon and mosambi were used as control in the experiment for the quantification of Vitamin C. All the three used methods were compared for their efficiency in vitamin C quantification. Quantification results are summarized in Table1. Result showed all the three methods are at par for quantification of vitamin C. Vitamin C content was found to be in the range of 0.78mg/g (by HPLC) to 0.87 mg/g using enzymatic method from fresh leaves of Moringa oleifera (Table1). Vitamin C content of Moringa oleifera leaves was almost two times as compared to Vitamin C content of lemon (0.42-0.54mg/g) and more than three times as compare to vitamin C content of mosambi (0.26 - 0.37 mg/g).

These results clearly demonstrated that Moringa oleifera leaves are rich source of vitamin C even in comparison to the well known vitamin C rich fruits lemon and mosambi. However, our results are not in line with earlier findings of Donovan, 2007 who reported that Moringa oleifera contains 7 times more vitamin C in comparison to oranges. Dried leaves of Moringa were found to contain 0.07- 0.14mg/g of vitamin C by different methods, which was much lower in comparison to fresh leaves. This reduction of vitamin C content after drying may be because of oxidative decomposition of vitamin C catalyzed by heat (Okere and Alonge, 2006; Shameen et al., 2009; Bineesh et al., 2005). It indicates that drying process is adversely affecting the amount of vitamin C. Stem samples of Moringa oleifera was found to contain 0.10-0.15mg/g of vitamin C using different methods which was much lower in comparison to fresh leaves. The values obtained in this study shows that Moringa could provide more allowance of vitamin C when compared with citrus fruits (lemon and Mosambi). This lends support to earlier findings of the use of Moringa plant as a rich source of vitamin C. This plant has an added advantage of other nutrients also. Considering these properties the cultivation and extraction of Moringa for ascorbic acid should be encouraged especially in developing countries where people are suffering from malnutrition. Research showed that this Miracle tree is probably the underutilized tree as far as its nutritional importance is concerned.

ACKNOWLEDGEMENTS

The authors would like to thank Prof. P. K. Khosla, Hon'ble Vice-Chancellor, Shoolini University of Biotechnology and Management Sciences, Solan and Foundation of Life Sciences and Business Management (FLSBM), Solan for providing financial support and necessary facilities for carrying out research. We are also indebted to Prof. D. R. Sharma for critical review of the paper and helpful comments.

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APPENDICES

Table 1: Quantification of Vitamin C in Moringa Collected from District Solan of Himachal Pradesh by Different Method

Sr. No.	Sample	Amount of Vit C by Titration Method(mg/g)	Amount of VitC by Spectropho to Meter Method (mg/g)	Amount of VitC by HPLC Method(mg/g)
1.	Dry Leaves	0.14	0.11	0.07
2.	Fresh Leaves	0.81	0.87	0.78
3.	Stem	0.13	0.15	0.10
4.	Lemon	0.47	0.54	0.42
5.	Mosambi	0.29	0.37	0.26